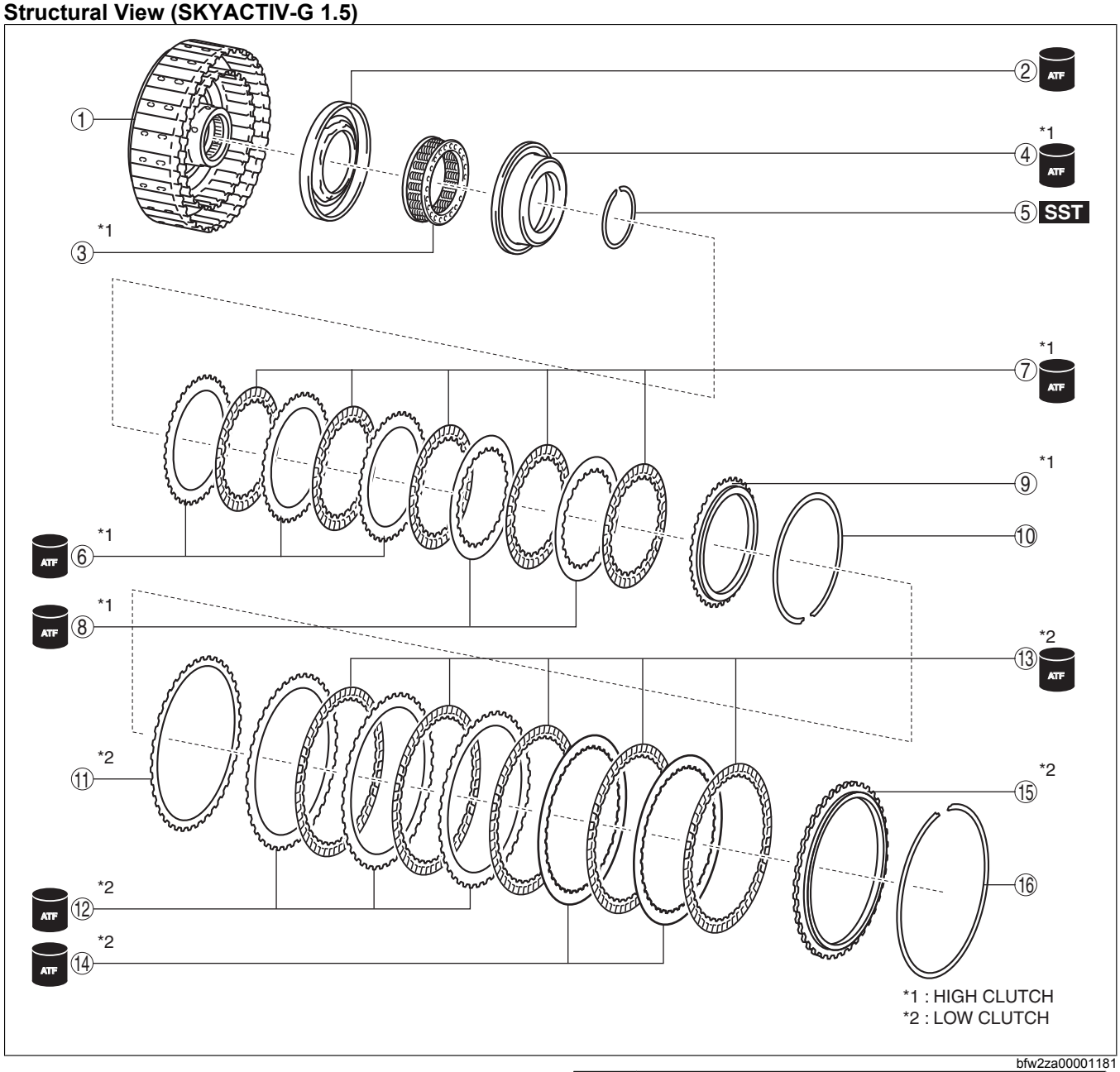
HIGH CLUTCH COMPONENT AND LOW CLUTCH COMPONENT ASSEMBLY [FW6A-EL/FW6AX-EL]

id0517006658z3

Structural View (SKYACTIV-G 1.5)

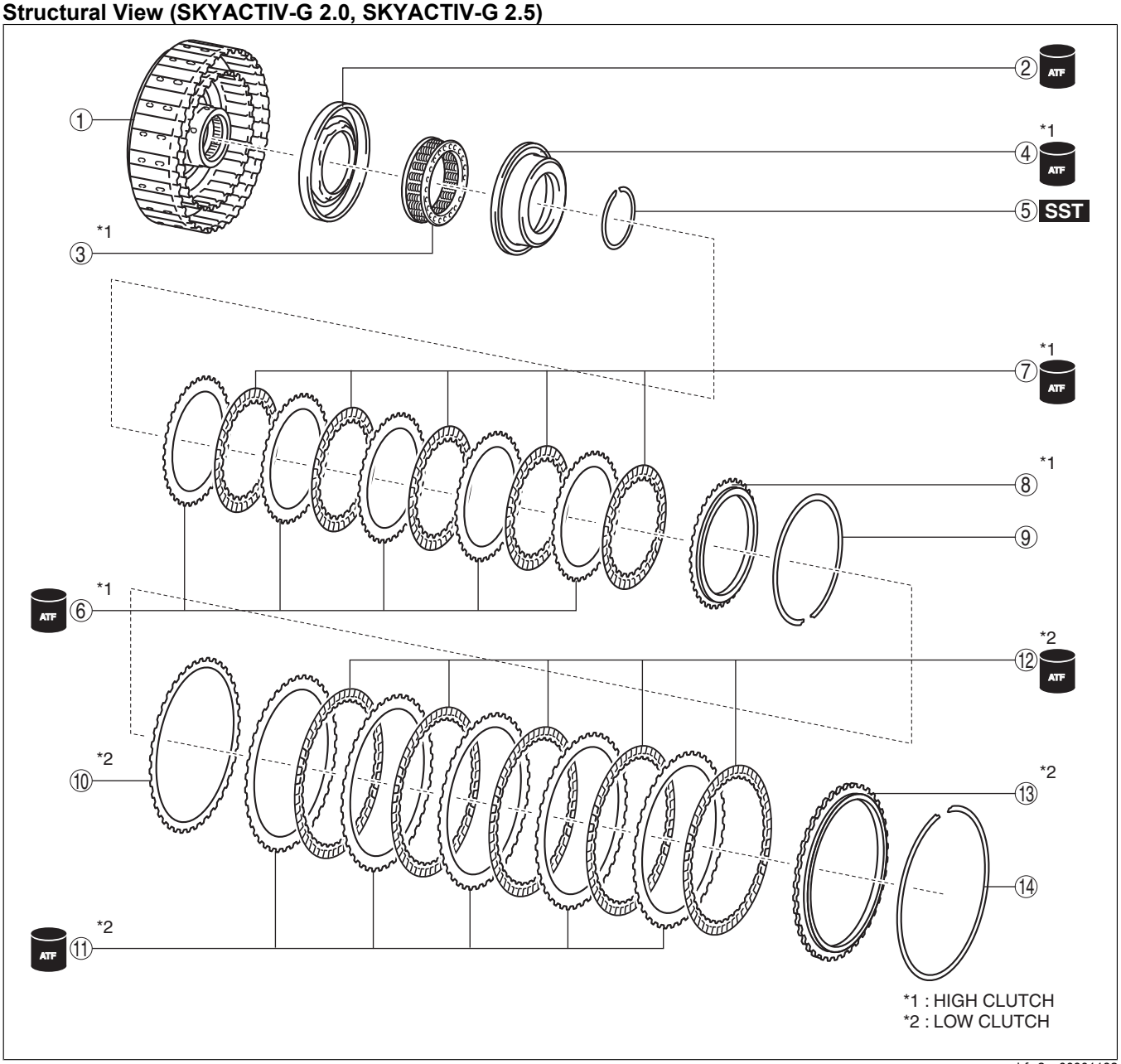


bfw2za00001181

1	High clutch drum and low clutch drum component
2	High clutch piston
3	Springs and retainer component (outer diameter approx. 79.5 mm {3.13 in})
4	Seal plate
5	Snap ring (outer diameter approx. 56.5 mm {2.22 in})
6	Driven plate (outer circumference engagement type) (inner diameter approx. 95.5 mm {3.76 in})
7	Drive plate (outer diameter approx. 116.8 mm {4.598 in})
8	Driven plate (inner circumference engagement type) (outer diameter approx. 116.8 mm {4.598 in})
9	Retaining plate (inner diameter approx. 95.5 mm {3.76 in})

10	Snap ring (outer diameter approx. 127.4 mm {5.016 in}) (selection)
11	Wave spring
12	Driven plate (outer circumference engagement type) (inner diameter approx. 139.9 mm {5.508 in})
13	Drive plate (outer diameter approx. 159.1 mm {6.264 in})
14	Driven plate (inner circumference engagement type) (outer diameter approx. 159.1 mm {6.264 in})
15	Retaining plate (inner diameter approx. 139.9 mm {5.508 in})
16	Snap ring (outer diameter approx. 169.3 mm {6.665 in}) (selection)

Structural View (SKYACTIV-G 2.0, SKYACTIV-G 2.5)



bfw2za00001182

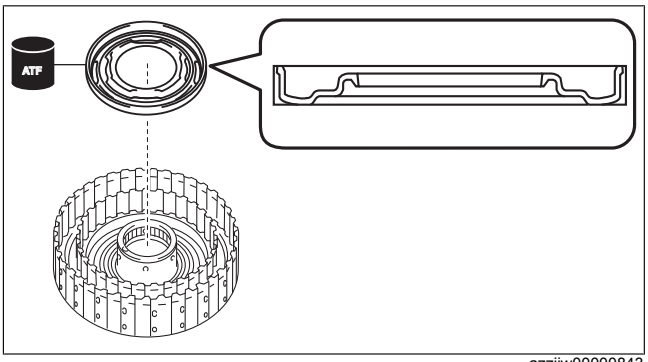
1	High clutch drum and low clutch drum component
2	High clutch piston
3	Springs and retainer component (outer diameter approx. 79.5 mm {3.13 in})
4	Seal plate
5	Snap ring (outer diameter approx. 56.5 mm {2.22 in})
6	Driven plate (inner diameter approx. 95.5 mm {3.76 in})
7	Drive plate (outer diameter approx. 116.8 mm {4.598 in})
8	Retaining plate (inner diameter approx. 95.5 mm {3.76 in})

9	Snap ring (outer diameter approx. 127.4 mm {5.016 in}) (selection)
10	Wave spring
11	Driven plate (inner diameter approx. 139.9 mm {5.508 in})
12	Drive plate (outer diameter approx. 159.1 mm {6.264 in})
13	Retaining plate (inner diameter approx. 139.9 mm {5.508 in})
14	Snap ring (outer diameter approx. 169.3 mm {6.665 in}) (selection)

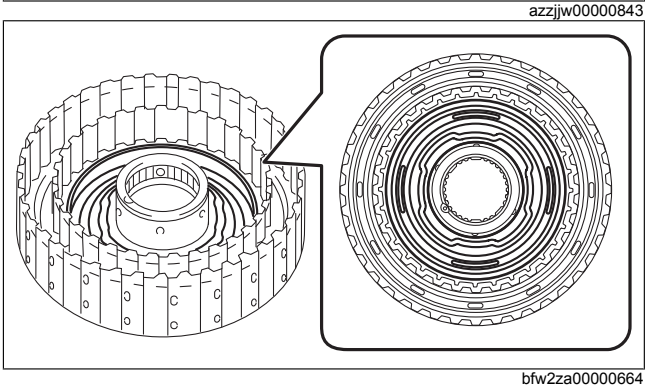
Assembly Procedure

1. Assemble the high clutch piston using the following procedure:
   (1) Apply ATF (ATF FZ) to the high clutch piston lip.

(2) Assemble the high clutch piston.



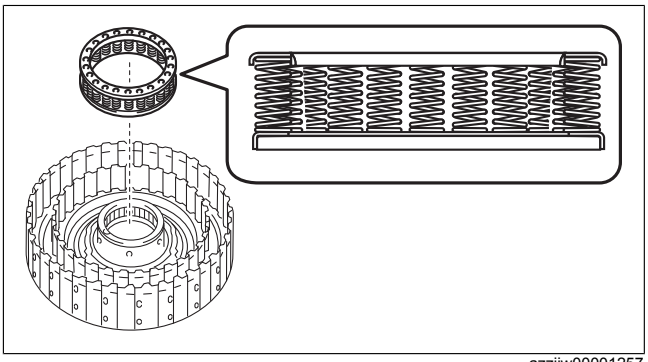
azzjjw00000843



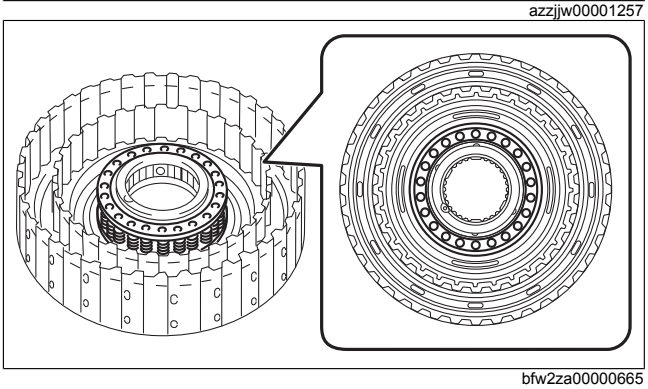
bfw2za00000664

2. Assemble the springs and retainer component.

Note
- Springs and retainer component size: Outer diameter approx. 79.5 mm {3.13 in}



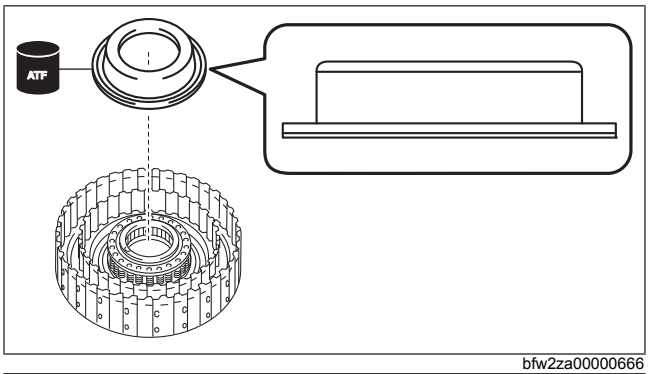
azzjjw00001257



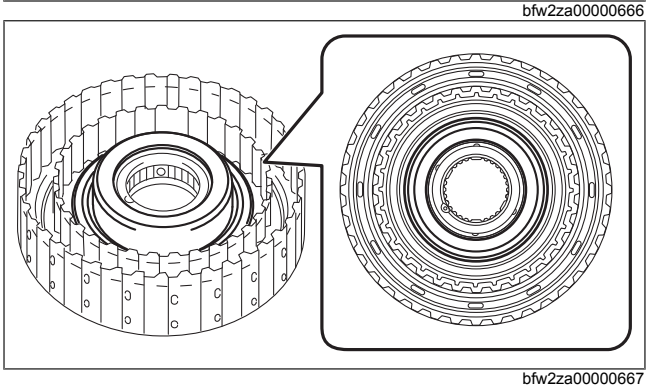
bfw2za00000665

3. Assemble the seal plate using the following procedure:
(1) Apply ATF (ATF FZ) to the seal plate lip.

(2) Assemble the seal plate.



bfw2za00000666

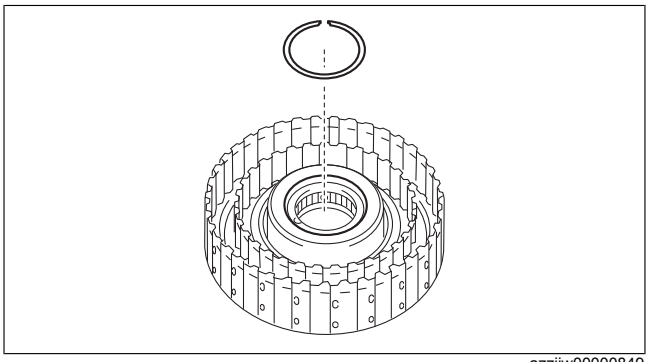


bfw2za00000667

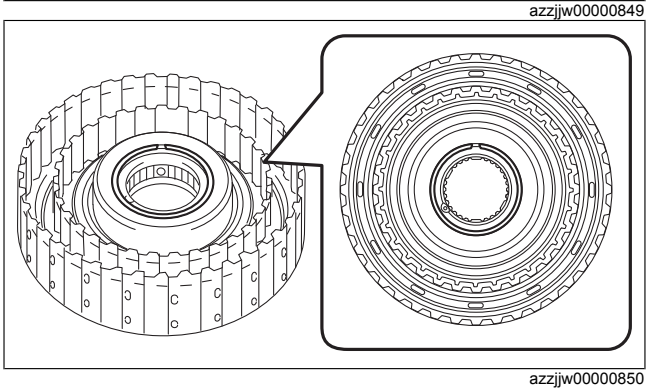
4. Assemble the snap ring using the following procedure:

Note
- Snap ring size: Outer diameter approx. 56.5 mm {2.22 in}

(1) Set the snap ring to the top of the seal plate.



azzjjw00000849



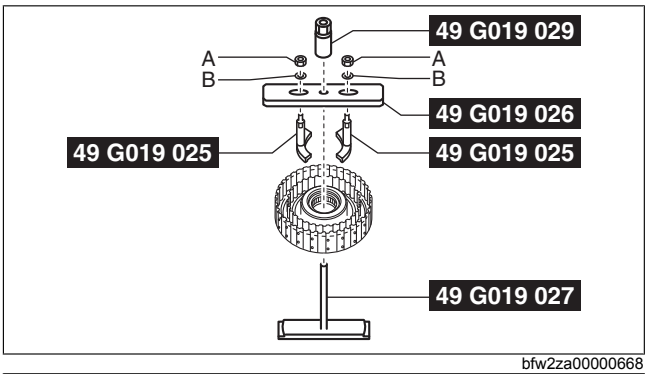
azzjjw00000850

(2) Install the SSTs.

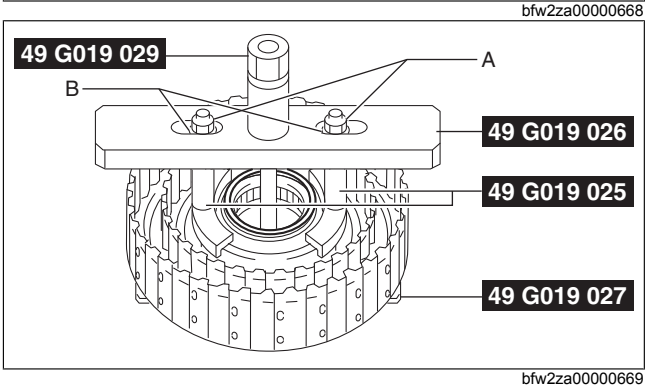
Note
- When installing the SST (49 G019 025) to the SST (49 G019 026), use the nuts included with the SST (49 G019 025), or M8×1.25 nuts.

A : Nut included with SST (49 G019 025), or M8×1.25 nut
B : Washer

A : Nut included with SST (49 G019 025), or M8×1.25 nut
B : Washer



bfw2za00000668

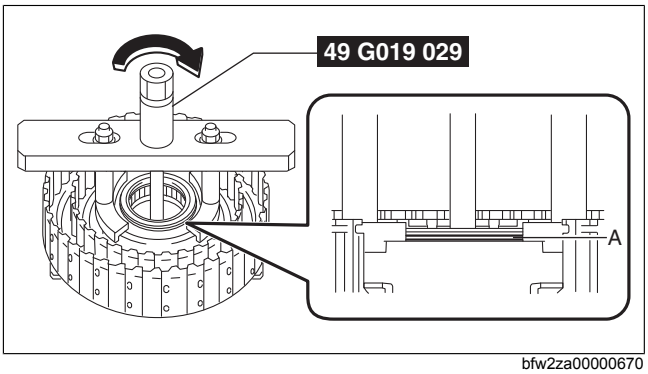


bfw2za00000669

(3) Tighten the SST (49 G019 029) until the snap ring groove of the high clutch drum and low clutch drum component comes out.

Caution
- **If the SST (49 G019 029) is tightened with excessive force, surrounding parts could be damaged. Stop tightening the SST when the snap ring groove of the high clutch drum and low clutch drum component comes out.**

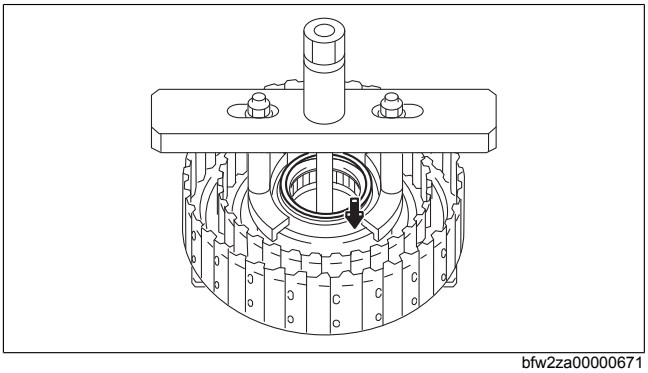
A : Snap ring groove



bfw2za00000670

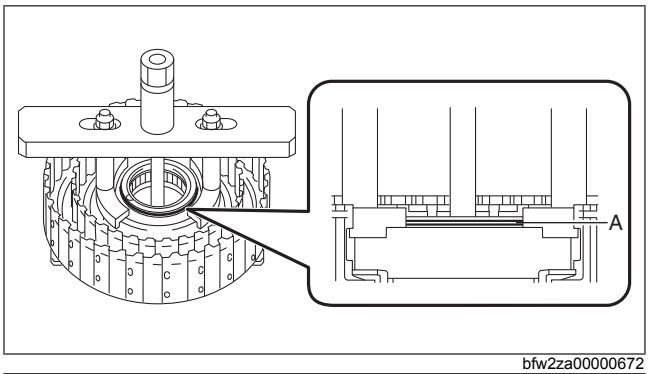
(4) Assemble the snap ring.

Caution
- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**



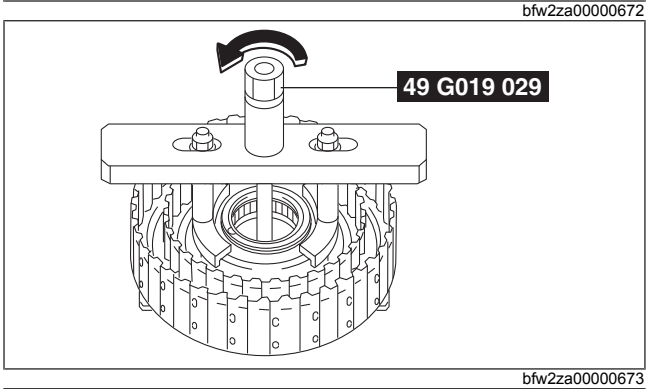
bfw2za00000671

A : Snap ring



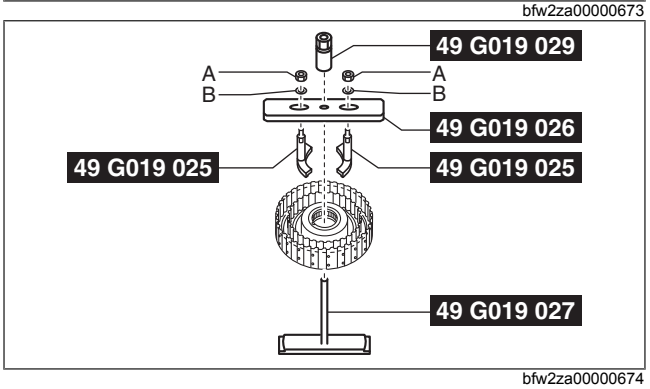
bfw2za00000672

(5) Loosen the SST (49 G019 029) and remove the SSTs.



bfw2za00000673

A : Nut included with SST (49 G019 025), or M8×1.25 nut
B : Washer



bfw2za00000674

5. Assemble the drive plates and driven plates using the following procedure: (SKYACTIV-G 1.5)

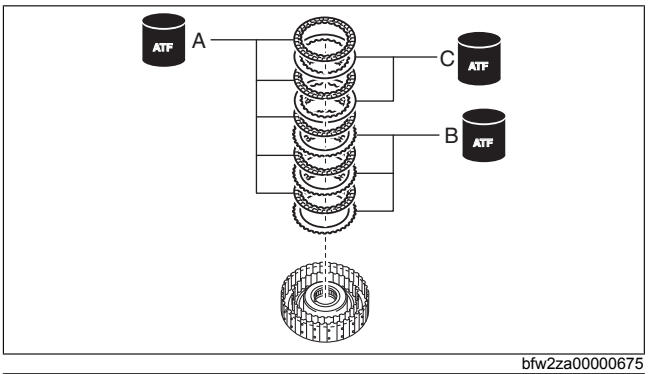
Note
- Drive plate size: Outer diameter approx. 116.8 mm {4.598 in}
- Driven plate size (Outer circumference engagement type): Inner diameter approx. 95.5 mm {3.76 in}
- Driven plate size (Inner circumference engagement type): Outer diameter approx. 116.8 mm {4.598 in}

(1) Apply ATF (ATF FZ) to the drive plates and driven plates.

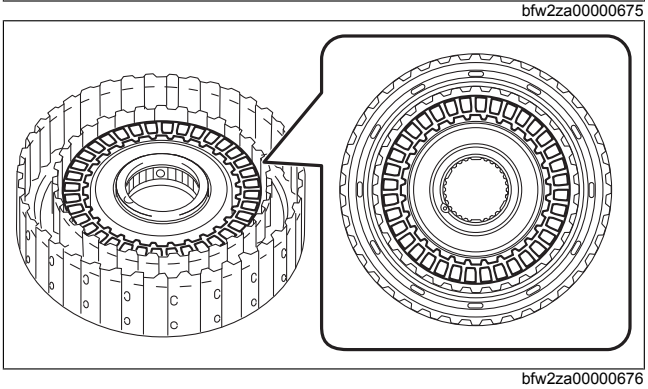
Caution
- **If the drive plate is replaced with a new one, immerse it in ATF (ATF FZ) for 2 hours or more to permeate the facing with ATF.**

(2) Assemble the drive plates and driven plates.
Assembly order
Driven plate (outer circumference engagement type)—drive plate—driven plate (outer circumference engagement type)—drive plate—driven plate (outer circumference engagement type)—drive plate—driven plate (inner circumference engagement type)—drive plate—driven plate (inner circumference engagement type)—drive plate

A : Drive plate
B : Driven plate (outer circumference engagement type)
C : Driven plate (inner circumference engagement type)



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bfw2za00000676

6. Assemble the drive plates and driven plates using the following procedure: (SKYACTIV-G 2.0, SKYACTIV-G 2.5)

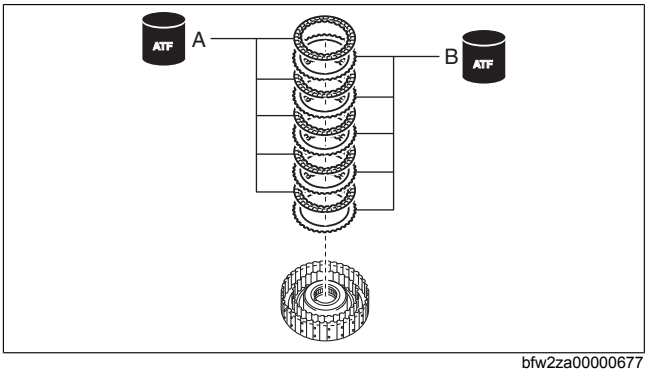
Note
- Drive plate size: Outer diameter approx. 116.8 mm {4.598 in}
- Driven plate size: Inner diameter approx. 95.5 mm {3.76 in}

(1) Apply ATF (ATF FZ) to the drive plates and driven plates.

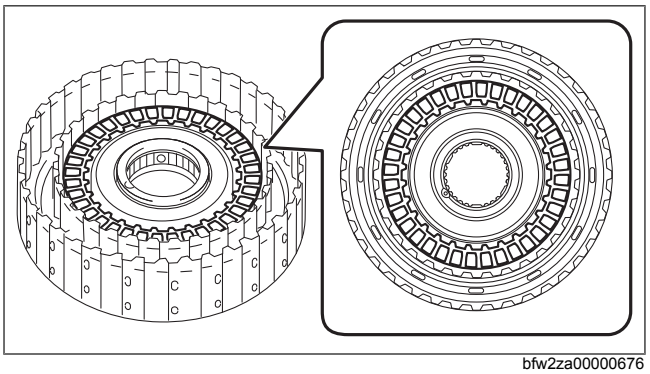
Caution
- **If the drive plate is replaced with a new one, immerse it in ATF (ATF FZ) for 2 hours or more to permeate the facing with ATF.**

(2) Assemble the drive plates and driven plates.
Assembly order
Driven plate—drive plate—driven plate—drive plate—driven plate—drive plate—driven plate—drive plate—driven plate—drive plate

A : Drive plate
B : Driven plate

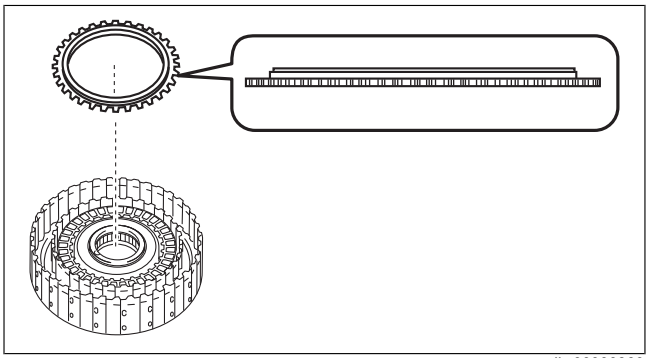


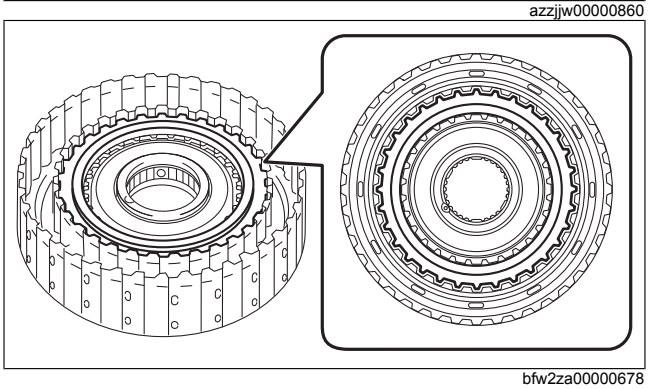
bfw2za00000677


bfw2za00000676

7. Assemble the retaining plate.

Note
- Retaining plate size: Inner diameter approx. 95.5 mm {3.76 in}


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bfw2za00000678

8. Assemble the snap ring using the following procedure:

Note
- Snap ring size: Outer diameter approx. 127.4 mm {5.016 in}

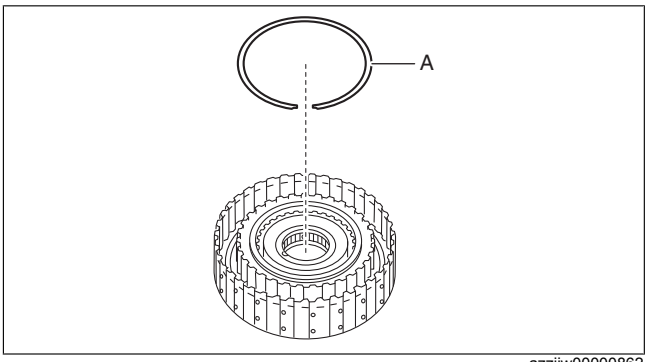
(1) Measure the high clutch clearance and select the appropriate snap ring. (See HIGH CLUTCH CLEARANCE MEASUREMENT/ADJUSTMENT.)

Note
- If the snap ring is assembled for the high clutch clearance measurement/adjustment, the following snap ring assembly procedure is not necessary.

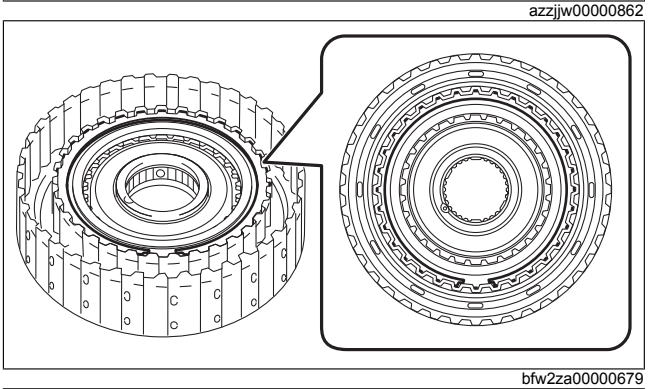
(2) Assemble the selected snap ring in Step (1).

Caution
- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**

A : Selection

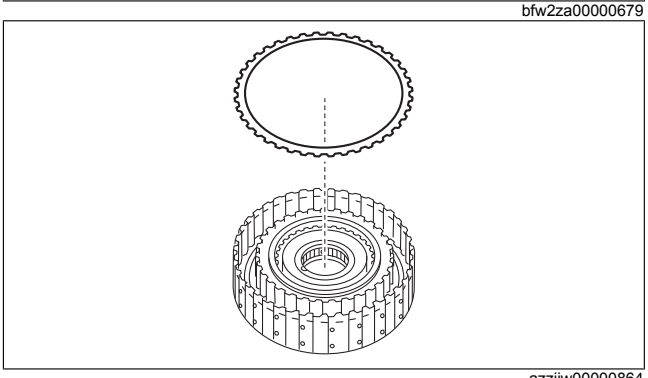


azzjjw00000862

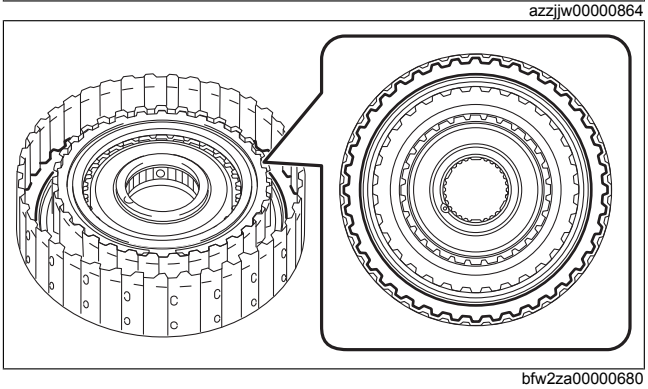


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9. Assemble the wave spring.



azzjjw00000864



bfw2za00000680

10. Assemble the drive plates and driven plates using the following procedure: (SKYACTIV-G 1.5)

Note
- Drive plate size: Outer diameter approx. 159.1 mm {6.264 in}
- Driven plate size (outer circumference engagement type): Inner diameter approx. 139.9 mm {5.508 in}
- Driven plate size (inner circumference engagement type): Outer diameter approx. 159.1 mm {6.264 in}

(1) Apply ATF (ATF FZ) to the drive plates and driven plates.

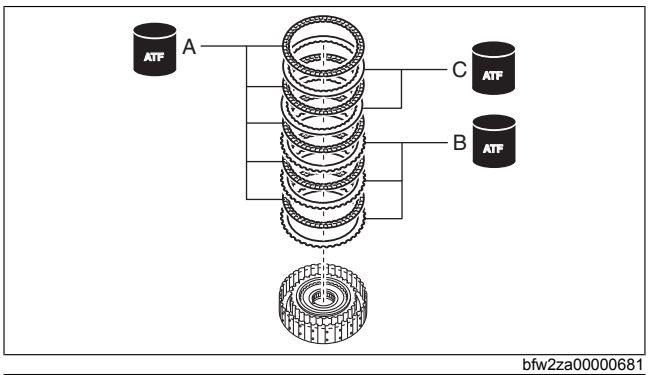
Caution
- **If the drive plate is replaced with a new one, immerse it in ATF (ATF FZ) for 2 hours or more to permeate the facing with ATF.**

(2) Assemble the drive plates and driven plates.

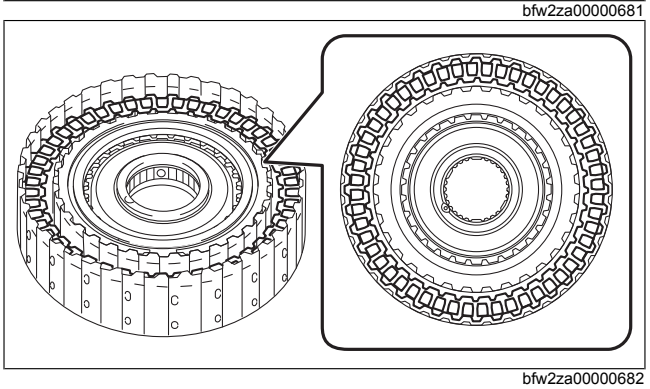
Assembly order

Driven plate (outer circumference engagement type)—drive plate—driven plate (outer circumference engagement type)—drive plate—driven plate (outer circumference engagement type)—drive plate—driven plate (inner circumference engagement type)—drive plate—driven plate (inner circumference engagement type)—drive plate

A : Drive plate
B : Driven plate (outer circumference engagement type)
C : Driven plate (inner circumference engagement type)



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bfw2za00000682

11. Assemble the drive plates and driven plates using the following procedure: (SKYACTIV-G 2.0, SKYACTIV-G 2.5)

Note

- Drive plate size: Outer diameter approx. 159.1 mm {6.264 in}
- Driven plate size: Inner diameter approx. 139.9 mm {5.508 in}

(1) Apply ATF (ATF FZ) to the drive plates and driven plates.

Caution

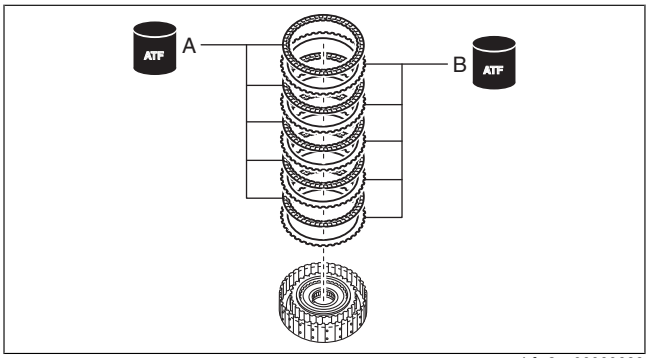
- **If the drive plate is replaced with a new one, immerse it in ATF (ATF FZ) for 2 hours or more to permeate the facing with ATF.**

(2) Assemble the drive plates and driven plates.

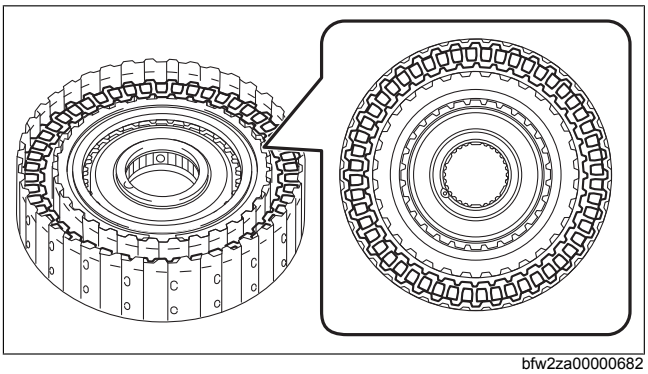
Assembly order

Driven plate—drive plate—driven plate—drive plate—driven plate—drive plate—driven plate—drive plate—driven plate—drive plate

A : Drive plate
B : Driven plate



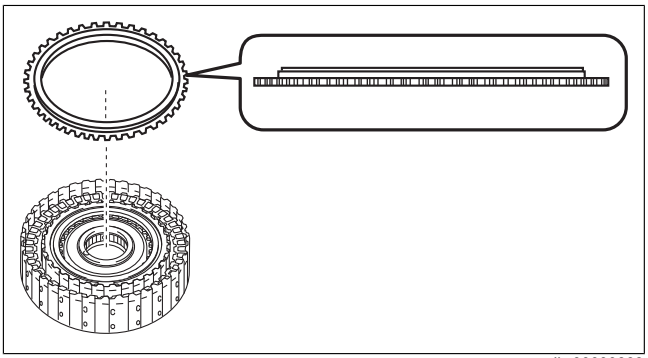
bfw2za00000683



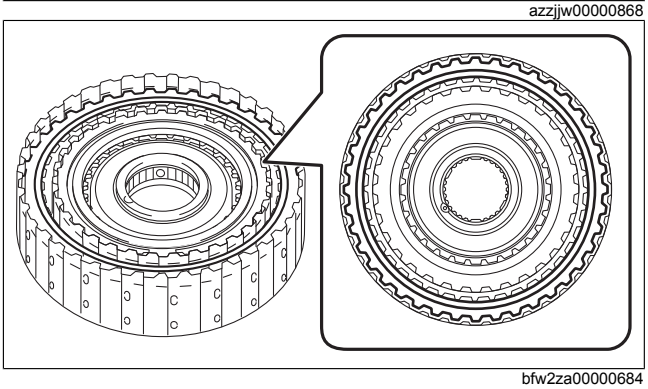
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12. Assemble the retaining plate.

Note
- Retaining plate size: Inner diameter approx. 139.9 mm {5.508 in}



azzjjw00000868



bfw2za00000684

13. Assemble the snap ring using the following procedure:

Note
- Snap ring size: Outer diameter approx. 169.3 mm {6.665 in}

(1) Measure the low clutch clearance and select the appropriate snap ring. (See LOW CLUTCH CLEARANCE MEASUREMENT/ADJUSTMENT.)

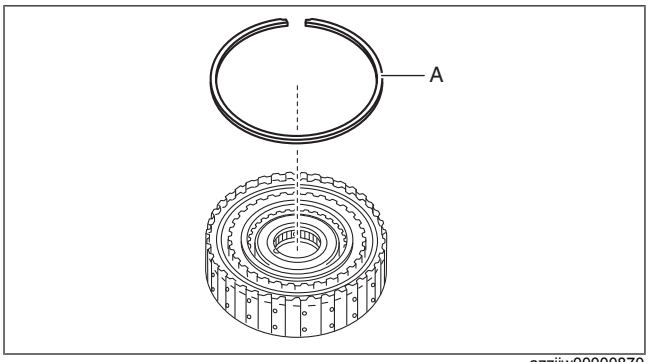
Note
- If the snap ring is assembled for the low clutch clearance measurement/adjustment, the following snap ring assembly procedure is not necessary.

(2) Assemble the selected snap ring in Step (1) to the position shown in the figure.

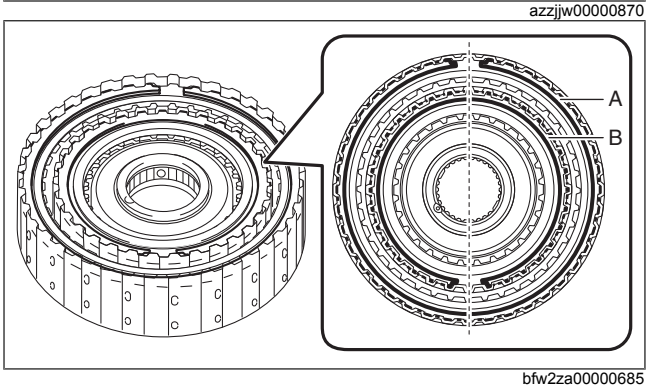
Caution
- **Assemble so that the end gap of the snap ring is positioned diagonally opposed to the end gap of the snap ring for the high clutch.**
- **After assembling the snap ring, verify that the snap ring is securely inserted into the bottom of the snap ring groove.**

A : Selection

A : Snap ring (low clutch)
B : Snap ring (high clutch)



azzjjw00000870

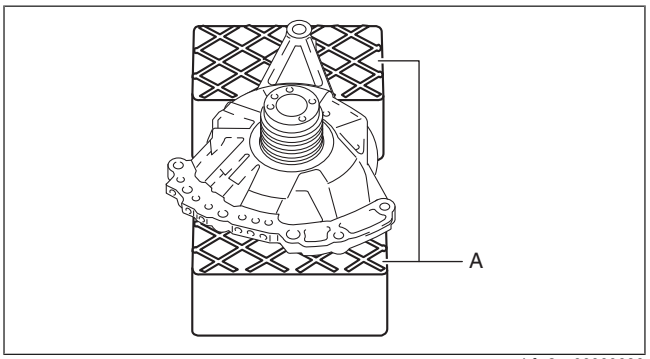


bfw2za00000685

14. Perform a simple inspection of the low clutch and high clutch using the following procedure:
    (1) Set the oil pump on the workbench as shown in the figure.

Note
- Using the rubber plates, adjust the alignment surface of the oil pump with the transaxle case so that it is level.

A : Rubber plate

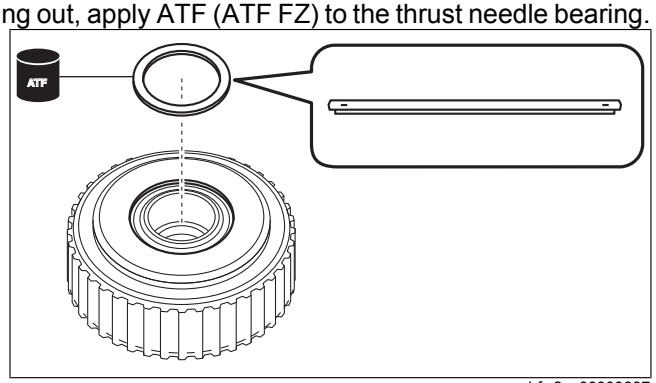


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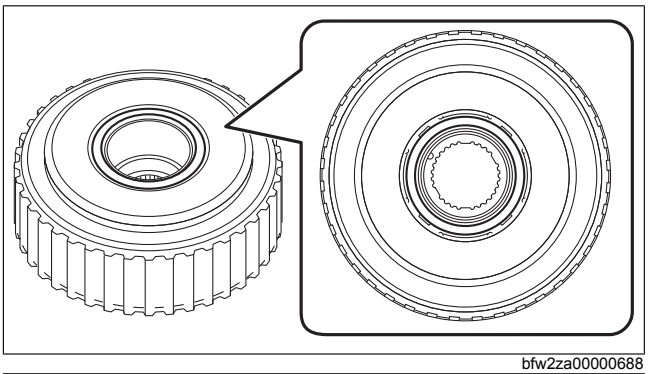
    (2) Assemble the thrust needle bearing to the high clutch component and low clutch component using the following procedure:

Note
- Thrust needle bearing size: Outer diameter approx. 72.7 mm {2.86 in}

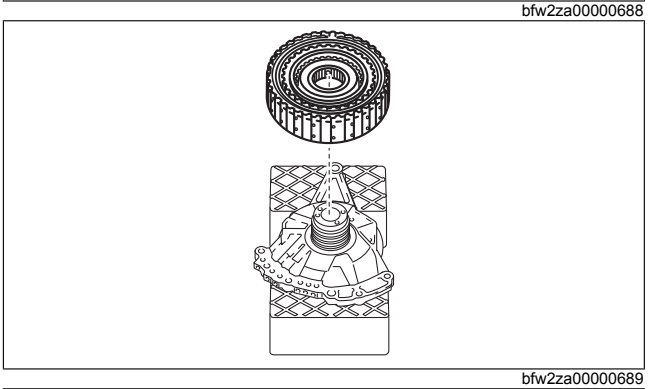
1) To prevent the thrust needle bearing from dropping out, apply ATF (ATF FZ) to the thrust needle bearing.
2) Assemble the thrust needle bearing.

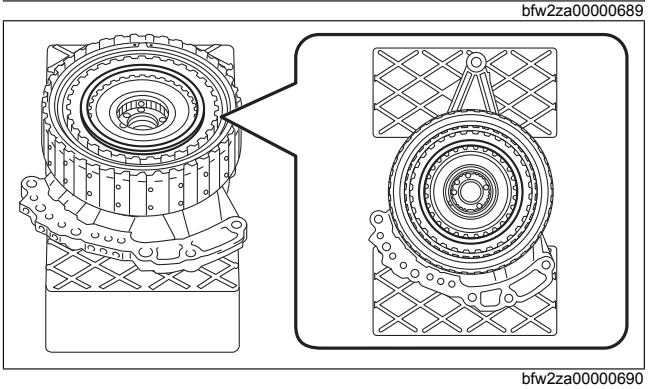


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(3) Assemble the parts assembled together in Step (2) to the oil pump.


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(4) Blow compressed air into the oil passage shown in the figure and verify the operation condition of the low clutch.

Warning

- **Always wear protective eye wear when using the air compressor. Otherwise, ATF or dirt particles blown off by the air compressor could get into the eyes.**

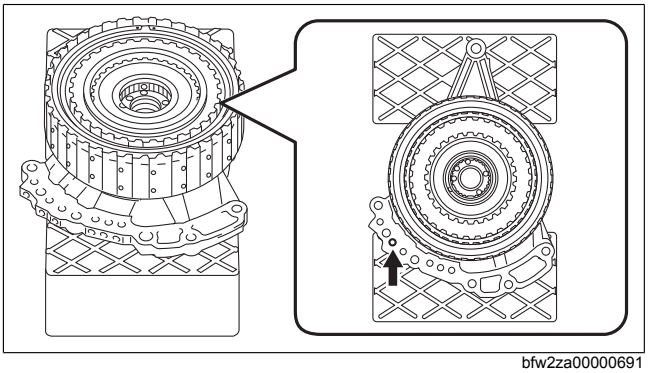
Caution

- **To prevent damage to parts, always use an air compressor which is adjusted to the indicated pressure.**

Compressed air pressure

0.39—0.44 MPa {4.0—4.4 kgf/cm^2, 57—63 psi}

- If there is a malfunction, perform disassembly again, verify the cause and repair the applicable part. (See HIGH CLUTCH COMPONENT AND LOW CLUTCH COMPONENT DISASSEMBLY [FW6A-EL/FW6AX-EL].)


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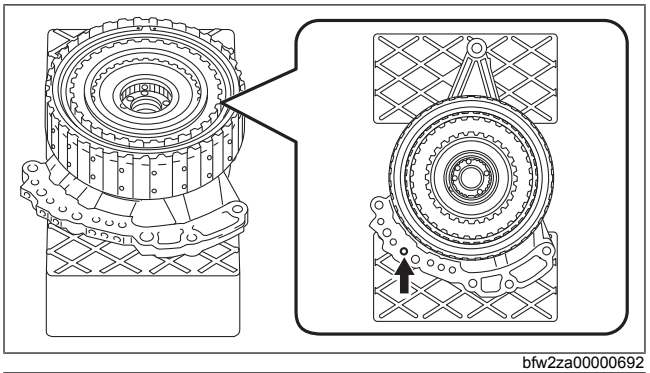
(5) Blow compressed air into the oil passage shown in the figure and verify the operation condition of the high clutch.

Warning
- **Always wear protective eye wear when using the air compressor. Otherwise, ATF or dirt particles blown off by the air compressor could get into the eyes.**

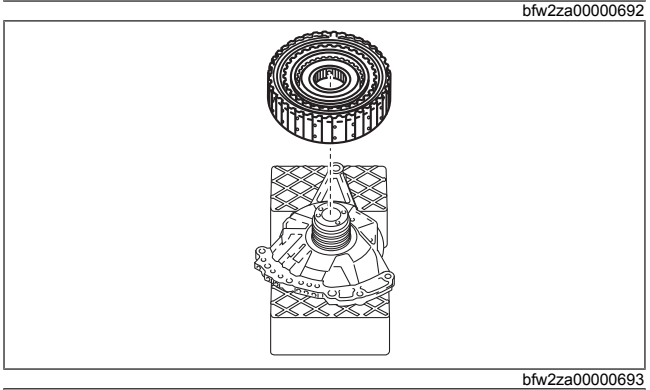
Caution
- **To prevent damage to parts, always use an air compressor which is adjusted to the indicated pressure.**

Compressed air pressure
0.39—0.44 MPa {4.0—4.4 kgf/cm^2, 57—63 psi}

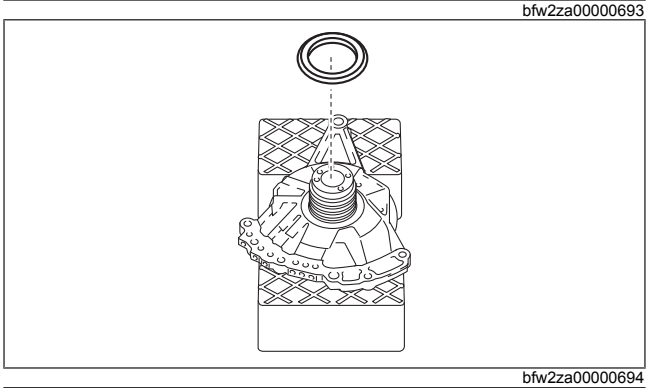
- If there is a malfunction, perform disassembly again, verify the cause and repair the applicable part. (See HIGH CLUTCH COMPONENT AND LOW CLUTCH COMPONENT DISASSEMBLY [FW6A-EL/FW6AX-EL].)


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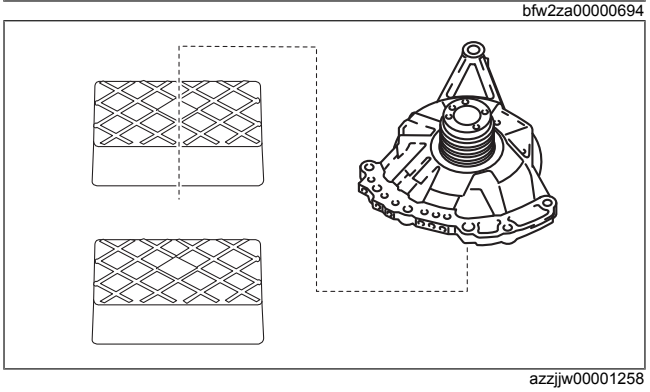
(6) Remove the high clutch component and low clutch component.


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(7) Remove the thrust needle bearing.


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(8) Take the oil pump off the rubber plates.


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